BorrowBox — Commands & Developer Setup Guide

Full-Stack Equipment Rental Marketplace | Developer Reference Manual

BorrowBox is a modern Java 25 + Spring Boot 3.5 equipment rental marketplace. This document lists all terminal commands for building, running, testing, configuring the database, and testing REST APIs.

1. Backend Development & Execution Commands

• Start Backend Development Server

Starts Spring Boot application on http://localhost:8080. Automatically runs Flyway migrations.

```
cd backend
mvn clean spring-boot:run
```

• Run Automated Unit & Concurrency Tests

Executes JUnit 5 test suite verifying PricingEngine and BookingConcurrency locks.

```
cd backend
mvn test
```

• Build & Package Executable JAR File

Compiles source code and packages executable JAR into backend/target/.

```
cd backend
mvn clean package -DskipTests
```

• Launch Packaged Production JAR

Runs compiled Spring Boot standalone application directly.

```
java -jar backend/target/borrowbox-backend-1.0.0.jar
```

2. Database Setup & Environment Configuration

BorrowBox automatically defaults to an in-memory **H2 Database** in MySQL mode if MySQL Server is offline. To connect to your local MySQL 8.x database server, configure these environment variables in PowerShell:

```
$env:DB_URL="jdbc:mysql://localhost:3306/borrowbox_db?useSSL=false&allowPublicKeyRetrieval;=true&serv
erTimezone;=UTC"
$env:DB_USERNAME="root"
$env:DB_PASSWORD="your_mysql_password"
```

• MySQL Database Creation Query:

```
CREATE DATABASE borrowbox_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
```

3. Documented Demo User Accounts

Role	Email Address	Phone Number	Password
Administrator	admin@borrowbox.com	+91 98765 00001	Password123!
Equipment Owner 1	rahul.owner@borrowbox.com	+91 98765 11111	Password123!
Equipment Owner 2	priya.owner@borrowbox.com	+91 98765 22222	Password123!
Renter 1	alex.renter@borrowbox.com	+91 98765 33333	Password123!

Renter 2	sneha.renter@borrowbox.com	+91 98765 44444	Password123!

4. Useful REST API cURL Commands

• Login API Endpoint (JWT Auth)

```
curl -X POST http://localhost:8080/api/auth/login \
-H "Content-Type: application/json" \
-d '{"email":"admin@borrowbox.com","password":"Password123!"}'
```

• Fetch Equipment Catalog

```
curl -X GET http://localhost:8080/api/equipment
```

• Fetch Categories List

```
curl -X GET http://localhost:8080/api/categories
```

• Integrated Web Application URL

```
http://localhost:8080
```

• H2 Web Console URL

```
http://localhost:8080/h2-console (JDBC URL: jdbc:h2:mem:borrowbox_db)
```